

Virtual-Coupling Operation for High-Speed Rail based on Following-Train Speed Profile Optimization

Bin Xu, Chaoxian Wu, Shaofeng Lu* and Fei Xue

Abstract—With the increasing railway transportation demands, railway operators need to enhance the capacity in high traffic corridors to improve transportation efficiency. The establishment of the concept “virtual coupling (VC)” provides an effective method for achieving higher capacity. The core idea is to reduce the train separation and to enable a maximized number of trains on the track. Based on the emerging modern communication techniques, the synchronous operation between different trains to form the coupled state can be achieved without any physical connection. In this paper, a following-train speed profile optimization model is proposed by employing Mixed Integer Linear Programming (MILP) approach to achieving VC with the leading train. Therefore, the safety constraints (i.e. moving block signaling (MBS)) are considered as the most important physical border between each train under the VC operation. Three cases under different scenarios are conducted on a generic high-speed rail, which proves the effectiveness of the proposed method on the speed profile optimization for following trains under the VC operation. It also shows the factors influencing the time headway under minimized-separation operations.

I. INTRODUCTION

On major rail lines during peak hours, how to plan the operation of trains to achieve maximum capacity and achieve the highest transportation efficiency has attracted attention among rail transportation studies today. Studies have proved that the key point to obtain the maximum capacity is to minimize the train safety headway while ensuring safety [1].

In three generations of standards established by the European Train Control System (ETCS) [2], each advancement presented by each generation considerably reduces its train separation. Fig. 1 explains the three generations of signaling systems and the VC system, with their train separation reductions identified. The illustrations of some terminologies can be found in Table I.

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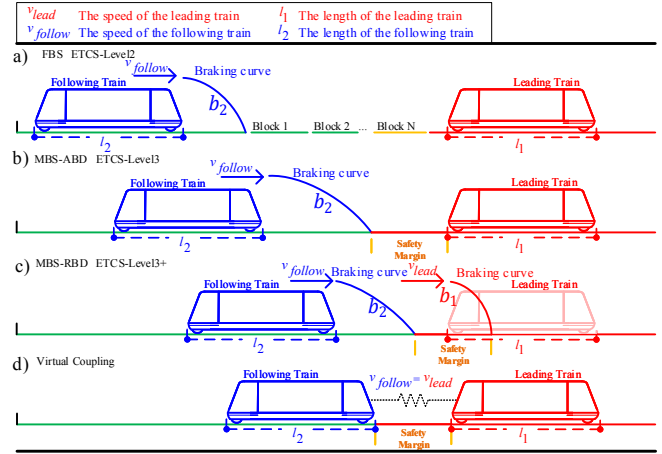


Fig. 1. The four generations of railway signaling system: where S_{train} is the train separation, d_s is the safety margin, b_1, b_2 is the braking distance of the leading/following train, a) fixed block signaling system; b) moving block signaling with an absolute braking distance ($S_{train} - b_2 \geq d_s$); c) moving block signaling with a relative braking distance ($S_{train} + b_1 - b_2 \geq d_s$); d) virtual coupling ($S_{train} \geq d_s$).

TABLE I
THE ILLUSTRATIONS OF DISTANCE SEGMENTS EXPRESSIONS

Terminology	Definition
Train Separation	The distance between the tail of leading train and the head of following train.
Safety Margin	A fixed safety separation in distance predefined by the rail system.
Safety Distance	The necessary train separation under safety constraints, i.e. FBS, MBS or VC
Margin Distance	The longest distance the following train could pass without collision under safety constraints.

- 1) Fig. 1-a) demonstrates ETCS level 2, also known as the fixed block signaling (FBS) system. In FBS, the rail track is recognized as multiple blocks which respectively limits its occupation. The Global System for Mobile Communications - Railway (GSM-R) transmits the accurate train profile to the radio block center (RBC), which is responsible for monitoring all the trains on the track and provides Movement Authority (MA) based on the data transmitted from each train [3]. Note that MA indicates the longest distance the train can travel. The MA in ETCS-2 is counted with the occupation status of each block predefined on the track.
- 2) Fig. 1-b) illustrates ETCS level 3, which is also called

MBS with an absolute braking distance (MBS-ABD). The train separation must be larger than a safety distance which consists of the braking distance of the following train and the safety margin. The MA provided by the RBC is no longer computed on the basis of fixed blocks but with the exact position and distance reported by each train.

- 3) Fig. 1-c) expresses the MBS with a relative braking distance (MBS-RBD). It joins the braking distance of the preceding train based on MBS-ABD, requiring that the two trains should always have a safety margin between their respective ends of braking distance. Duan *et al.* conducted a capacity analysis of the three signaling systems and examined their capacity under six scenarios. The results in [4] show that MBS-RBD can increase its transportation capacity by 59% compared with MBS-ABD.
- 4) Fig. 1-d) shows the mechanism of VC. In a VC train group, all trains can be regarded as a whole, which means they are virtually coupled together and move synchronously. Suppose all the characteristics of a train are in the attribute vector $A_i\{a_1, a_2, \dots, a_i\}$, including factors of the train line, operation direction, destination station, and other information. In VC status, the leading train attribute vector $A_{i,lead}$ and the following train attribute vector $A_{i,follow}$ should be equal to each other, i.e. $A_{i,lead} = A_{i,follow}$ [5].

The employment of VC concept enables the train to obtain their shortest separation distance. Rivera *et al.* pointed out in their study that the use of VC technology could reduce the headway by 66% compared with the traditional MBS-ABD [6]. But at the same time, due to different operating circumstances of trains, such as entering or leaving stations, encountering complex slopes, speed limit sections and weather conditions, etc., the virtually coupled train group is bound to go through a process of disturbance. Accordingly, it is necessary to adjust the operating status of the train from time to time to form the “coupled” train group and overcome disturbance, which purposed strict communication requirements to actualize VC.

In MBS system, each participating train needs to conduct radio-based Vehicle-to-Infrastructure (V2I) communication in real-time (usually within 1-5s) reporting its speed, position, and acceleration information [3]. The RBC provides feedback MAs based on the data transmitted through V2I to each train. For VC, the leading train on the track transmit its kinetic profile through Vehicle-to-vehicle (V2V) communications to the rear one [7]. The following train in the group considers the leading train’s kinetic information and make a decision combining with the MA provided by the RBC. The train communication topology could be found in Fig. 2.

When the train operation encounters interference which leads to coupling or decoupling, the coupled structure would be disrupted. An effective safety mechanism will be required to ensure that the leading and following trains could avoid collisions. Many studies have discussed detailed models to ensure safe operation using VC technology. Di Meo *et*

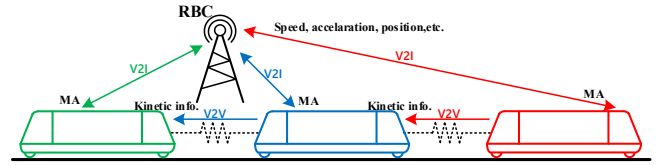


Fig. 2. The VC communication topology: through V2V, the leading trains transmit their kinetic info. (i.e. speed and position); through V2I, the train reports their operation parameters to RBC, while RBC provides feedback MAs to the train [3][7].

al. established a coupling operation method based on the ETCS framework, taking into account numerous scenarios to evaluate its flexibility [8]. The VC structure proposed by Quaglietta *et al.* combines the operation mechanism of ETCS-3 with V2V communication [3]. Xun *et al.* discussed the speed profile of coupled trains and added speed limit control to prevent overspeeding [9]. Park *et al.* designed a gap controller based on sliding mode control to manage the distance between the leading and following trains under an error correction scheme [10]. Chen *et al.* proposed a coordinated collision mitigation approach with model predictive control and organized a numerical analysis on emergency braking of both homogeneous and heterogeneous fleets [11]. In this research, VC is considered based on MBS-RBD safety constraints to ensure its safety. The model will evaluate the braking distances of both the leading and following trains, as well as the safety margin to achieve the MA.

This research proposes a linear model based on MILP to optimize the full speed profile of the following train with the given information of the leading train. The case study in the paper shows the effectiveness and robustness of the proposed method in different operation scenarios. In addition, the performance of the VC operation under separation minimizing is also discussed in the paper based on the case study results. The contributions of this article are summarized as follows:

- An MILP model is formulated to realize the following train speed profile optimization with the objective function to directly minimize the distance between each train under the constraint of MBS-RBD.
- The case study for scenarios of the leading train with or without disturbance is given, which shows the shortest train separation can be achieved under VC by using the proposed method.
- The factors influencing the railway headway are analyzed after applying the model on a series of tests.

The remainder of the paper is organized as follows: section II proposes the mathematical model in detail including train motion, energy flow, interactive approach and the modeling objective. Section III designs a rail line with three scenarios to examine the performance of the VC strategy combined with MBS-RBD. Section IV concludes the research so far and discusses future directions.

II. MATHEMATICAL MODELING

A. Single Train Motion Analysis with Energy Flow

In this MILP model, the total scheduled operation carries a time cost of T arranged between two stations is divided equally into N time segments Δt . The whole track length D along the railway line is separated into N distance segments Δd_i , each of them carries a time cost of Δt . Their relationships are shown in (1) and (2).

$$N = \frac{T}{\Delta t} \quad (1)$$

$$\sum_{i=1}^N \Delta d_i = D \quad (2)$$

$N + 1$ speed points need to be optimized at each time point t_i . The optimized speed v_i could form the expected full track speed profile by linearly connected them along. In each of the time-segment, the acceleration motion could be considered uniformly. Considering passenger comfort and locomotive performance, the acceleration a_i should not exceed the maximum deceleration limitation $a_{max,de}$, and the maximum acceleration $a_{max,ac}$, expressed as (3).

$$\frac{v_{i+1} - v_i}{\Delta t} = a_i, a_i \in [a_{max,de}, a_{max,ac}] \quad (3)$$

Besides the acceleration, speed limitation $v_{i,lim}$ should be taken into account along the line. The train speed should not exceed the speed limits as (4).

$$v_i \leq v_{i,lim} \quad (4)$$

The average speed $v_{i,ave}$ and distance traveled in each time period Δd_i are computed respectively with the equations (5)(6).

$$v_{i,ave} = \frac{v_i + v_{i+1}}{2} \quad (5)$$

$$\Delta d_i = v_{i,ave} \times \Delta t \quad (6)$$

The Davis equation is used to represent the drag force $f_{i,drag}$ of the train along with Davis coefficients A, B and C, shown as (7).

$$f_{i,drag} = A + Bv_{i,ave} + Cv_{i,ave}^2 \quad (7)$$

In this model, the regenerative braking energy of the train is not considered, though it is not difficult to adapt the model to regenerative braking cases. The relative energy transformation of the train is considered as (8).

$$\frac{1}{2}M(v_{i+1}^2 - v_i^2) + f_{i,drag}\Delta d_i + Mg\Delta h_i \leq E_i \quad (8)$$

where E_i is the energy consumed by train traction systems in the i^{th} time segment, M is the mass of the train, g is the gravity coefficient and Δh_i is the altitude difference between i^{th} and $(i + 1)^{th}$ time segment. The gradient information is calculated in Section 2-B. The model also ensures the train operates under the limitation of maximum traction force F_{max} and the maximum traction power P_{max} . Eqs. (9) and (10) presents the constraints above.

$$0 \leq E_i \leq F_{max}\Delta d_i \quad (9)$$

$$0 \leq E_i \leq P_{max}\Delta t \quad (10)$$

B. Piecewise-linear Approach

In this research, the square of speed v^2 is used as the decision variables because it is directly linearly related to the energy transformation. Therefore, (3)-(7) exist nonlinear constraints. The Piecewise-linear (PWL) approach is introduced to address these constraints. PWL can divide nonlinear problems into several linear segments so that linearization methods can be used for optimization to mitigate the computation load. It can represent a non-negative function with a series of special ordered set type 2 (SOS2) where at most two adjacent variables could be nonzero. The sum of the SOS2 should be equal to 1 [12], which requires sets of integers to meet the goal during modeling. Two series of PWL section points are chosen to represent the whole range of the speed/distance in order to build the train motion and energy model. The product of SOS2 and PWL points indicates the target variable after linearization, their specific modeling constraints could be reference to [13], [14]. In the model, three sets of SOS2 are settled with the K sections, representing the speed v_i , the average speed $v_{i,ave}$, and the distance passed so far d_i respectively. The K section points are distributed evenly between the minimum value to maximum value for speed/distance.

Define a set of PWL speed points $V_{k,PWL}$, and a set of SOS2 $\alpha_{i,k}$ representing v_i^2 is expressed by (11).

$$v_i^2 = \sum_{k=1}^K V_{k,PWL} \times \alpha_{i,k} \quad (11)$$

The approximation of speed v_i is represented by v_i' according to (12).

$$v_i' = \sum_{k=1}^K V_{k,PWL} \times \alpha_{i,k} \quad (12)$$

Another set of SOS2 $\beta_{i,k}$ representing $v_{i,ave}$ is settled to help acquire the linearized approximation $v_{i,ave}'$, expressed by (13).

$$v_{i,ave}' = \sum_{k=1}^K V_{k,PWL} \times \beta_{i,k} \quad (13)$$

Besides the speed, (4) and (8) require altitude and speed limitation profile depends on the exact distance from the initial point d_i . The linearization process is the same as v_i which needs additional sets of PWL points $D_{k,PWL}$ and its SOS2 set $\gamma_{i,k}$. The distance passed d_i is determined by (14) and the SOS2 $\gamma_{i,k}$ should follows (15) as well.

$$d_i = \sum_{n=1}^i v_{n,ave}' \times \Delta t = \sum_{k=1}^K D_{k,PWL} \times \gamma_{i,k} \quad (14)$$

$$\sum_{k=1}^K \gamma_{i,k} = 1, 0 \leq \gamma_{i,k} \leq 1, i = 1, 2, 3 \dots (N + 1) \quad (15)$$

The altitude and speed limitation both require a new set of PWL sets, and we use $H_{k,PWL}$ to denote altitude and $L_{k,PWL}$ to denote speed limitation. (16) points out the

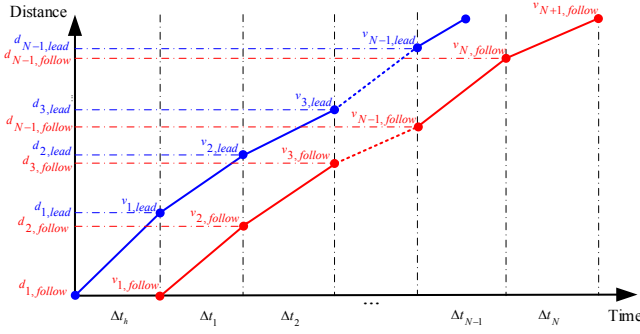


Fig. 3. The time, position and speed of both trains at each segment, where Δt_h is the time headway of the following train.

gradient difference in each segment while (17) models the speed limits.

$$\Delta h_i = \sum_{k=1}^K H_{k,PWL} \times \gamma_{i+1,k} - \sum_{k=1}^K H_{k,PWL} \times \gamma_{i,k} \quad (16)$$

$$v_{i,lim} = \sum_{k=1}^K L_{k,PWL} \times \gamma_{i,k} \quad (17)$$

C. Interactive Analysis and Objective Function

In the model, VC employs MBS-RBD to avoid collision hazards. As explained in Fig. 1, the “crash on the soft wall” system should follow the constraint (18), where the passed distance of the leading/ following train $d_{i,lead}$ and $d_{i,follow}$ are assumed to be the distance between their mass points and initial points, while the length of the leading train is l_1 and the length of the following train is l_2 . $b_{i,lead}$ stands for the braking distance of the leading train, $b_{i,follow}$ is the braking distance of the following train, and d_s is the safety margin. The demonstrations of the variables and operation schematic is shown in Fig. 3 and the braking distance is computed as (19).

$$[d_{i,lead} - (\frac{1}{2}l_1 + \frac{1}{2}l_2) - d_{i,follow}] + (b_{i,lead} - b_{i,follow}) \geq d_s \quad (18)$$

$$b_{i,lead} = \frac{v_{i,lead}^2}{2a_{max,de}}, b_{i,follow} = \frac{v_{i,follow}^2}{2a_{max,de}} \quad (19)$$

The purpose of VC is to reduce train separation. Based on this concept, simple minimization of the speed difference to form $v_{i,lead} = v_{i,follow}$ is not a feasible solution. In this research, the objective function is set to be directly minimizing the train separation, with its mathematical expression shown in (20). By applying the objective function, a VC status is realized with the minimized separation under safety constraints.

$$\min \sum_{i=1}^{N+1} (d_{i,lead} - (\frac{1}{2}l_1 + \frac{1}{2}l_2) - d_{i,follow}) \quad (20)$$

s.t. (1) - (19)

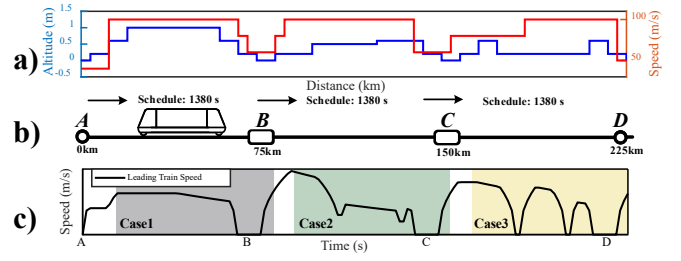


Fig. 4. a) The speed limits and altitude information on the track; b) The full track profile overview; c) The full journey speed profile of the leading train, where the grey window stands for case 1, the green window stands for case 2, and the yellow window stands for case 3.

TABLE II

THE PARAMETERS OF THE TRAIN (BASED ON MODEL CRH380A)

l_1, l_2 (m)	M (t)	P_{max} (kW)	F_{max} (kN)
203	442	8760	200
V_{max} (m/s)	Davis Coefficients		
97.22	A	B	C
a_{max} (m/s ²)	(kN)	(kN · s/m)	(kN · s ² /m ²)
1.2	5.6	0.036	0.00121

III. CASE STUDY AND DISCUSSION

To verify the effectiveness of train separation reduction and the performance of MBS-RBD safety mechanism under various situations, in this study the MILP model is tested under three scenarios with a single high-speed rail track. All the simulations are conducted with Gurobi® on MATLAB®. These cases are selected for the following train operation analysis, where the leading train is operating under various circumstances:

- Case 1: operates smoothly without disturbances.
- Case 2: encounters two slight disturbances.
- Case 3: encounters two severe disturbances.

While these disturbances may consists of the scenarios of sudden failures of the preceding vehicle, abnormal weather change, or improper operation of train passengers.

The three scenarios are assumed to be traveling on a high-speed rail line with a full journey length of 225 km and four stations A, B, C, D . The distance between each pair of the adjacent stations is 75 km and the scheduled time is 23 minutes, as shown in Fig. 4-b). When the train arrived at one station, it stops for another 207 s waiting for passengers to get on and off. The safety margin is set to be 1500 m. Moreover, this track has speed limits and altitude differences during the path as shown in Fig. 4-a). The speed profile of the leading train during the whole track is shown in Fig. 4-c), while the colored periods are the cases chosen to analyze for the train-following model. The parameters of the high-speed train operating on this track are displayed in Table II.

A. Case 1

In Case 1, the leading train travels from station A to station B according to the schedule requirements and continues to move forward to the next station after staying at station B

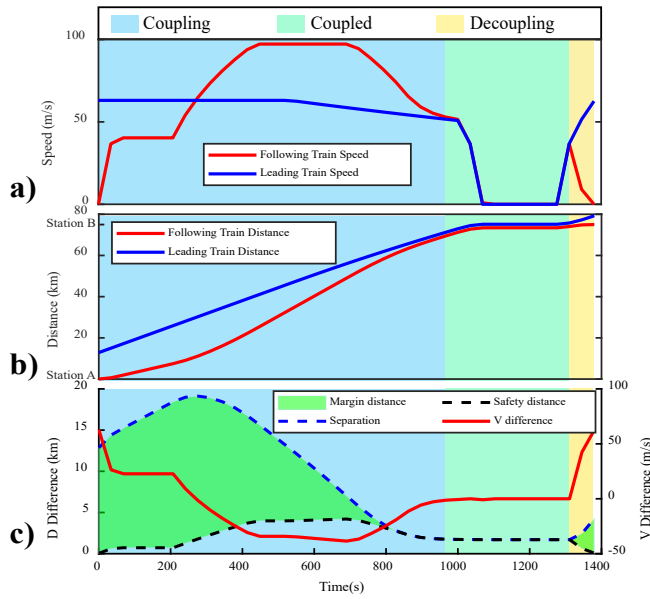


Fig. 5. The train speed profiles in case 1: a) the speed trajectory for the leading/following trains; b) the traversed distance paths for the leading/following trains; c) the margin distance of the following train and the speed difference during the period in case 1.

for 207 s. The following train leaves station A after 310 s of the leading train's departure. Their speed trajectories are marked accordingly in Fig. 5-a). The following train experienced a speed limitation of 40 m/s for the first 11 km and then accelerates to its maximum speed. Finally, its speed drops to follow the leading train and formed a coupled state. Their traversed distance are plotted under the same time scale in Fig. 5-b). The following train performs a chasing curve during the coupling period to get close to the leading train. The train separation and safety distance are plotted with dash lines in Fig. 5-c). Their interval, highlighted in green, shows the margin distance of the following train. The separation-optimized condition is that the separation equals to safety distance, which only happens under the VC operation with a safety distance equal to safety margin.

In this case, the following train chases the leading train with the maximum speed and acceleration to shorten the separation under speed restrictions and then decelerates to arrive at the next station. The two trains form a coupled state when the leading train is about to decelerate. The following train follows to stop while the leading train enters the station with a train separation of 1703 m, which is exactly $\frac{l_1}{2} + \frac{l_2}{2} + d_s$, the minimum safety distance. The leading train enters station A after the leading's departure and both trains at this stage decouple.

B. Case 2

In Case 2, the leading train meets two slight disturbance when traveling from station B to station C. The speed decreases to 30 m/s two times before reaching station C. The following train departs after staying at station A for 207 s, which forms a time headway of 310 s. The following train directly accelerates to its maximum speed and repeatedly

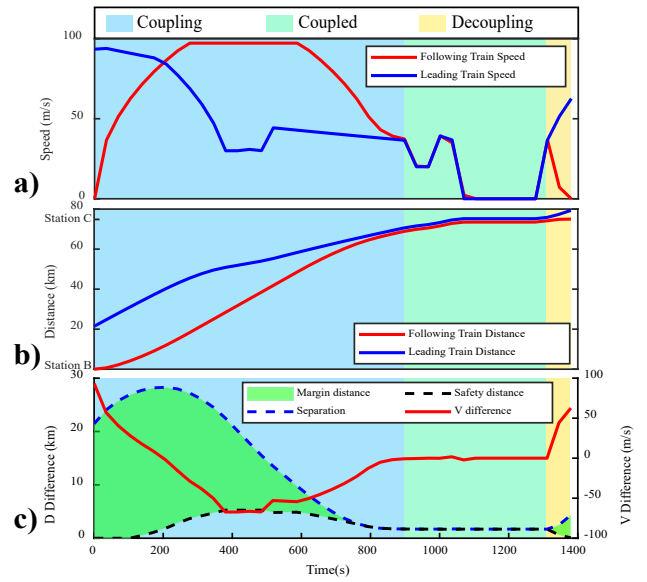


Fig. 6. The train speed profiles in case 2: a) the speed trajectory for the leading/following trains; b) the traversed distance paths for the leading/following trains; c) the margin distance of the following train and the speed difference during the period in case 2.

chasing the leading train, which can be found in Fig. 6-a). The train pair forms a coupled state before the leading train encounters the second disturbance. Fig. 6-b) displays the traversed distance of both trains. Similarly to case 1, the margin distance of the following train decreases and finally reaches zero after entering coupled state.

In this case, the following train ignores the first disturbance of the leading train since the margin distance is redundant. Before the second disturbance, the following train catches up with the leading train and goes to disturbance synchronously since there is no redundant margin distance that allows the following train to pass at the original speed. A virtually coupled status is formed. The following train stops at a minimum safety distance behind station C and waits for 207 s for the leading train to leave. The coupled pair decoupled when the following train decelerates.

C. Case 3

In Case 3, the leading train experiences two severe disturbance which is forced to stop two times. After 310 s of the leading train's departure, the following train accelerates under speed limits. Their speed and distance profile could be found from Fig. 7-a) and 7-b). Similar to the two previous cases, Fig. 7-c) shows the margin distance of the following train reaches zero before the second disturbance with a train separation equal to a minimum safety distance. In this case, the following train neglects the first severe disturbance and get affected by the second. The virtually coupled state is decoupled as the following train decelerates.

D. Minimum Time Headway Analysis

Based on the proposed method and above three scenarios, it is also found that minimizing the train separation could

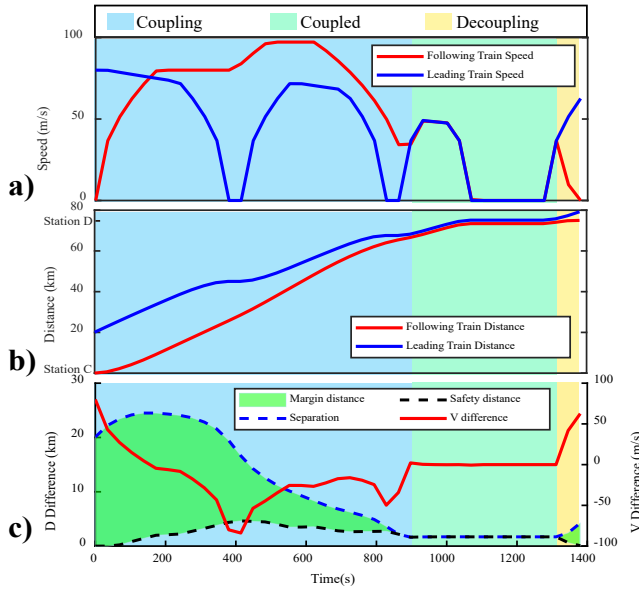


Fig. 7. The train speed profiles in case 3: a) the speed trajectory for the leading/following trains; b) the traversed distance paths for the leading/following trains; c) the margin distance of the following train and the speed difference during the period in case 3.

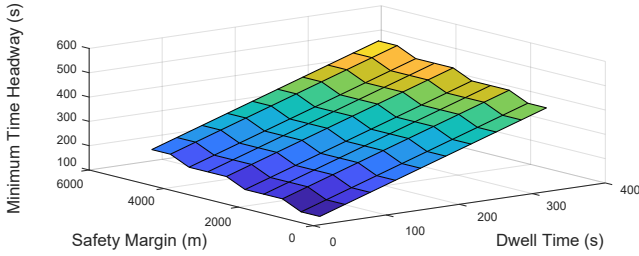


Fig. 8. The relationship between the leading train's dwell time, safety margin and minimum time headway between the leading train and the following train.

lead to VC operations. However, It is found from the case study that on a single track, minimizing real-time train separation can not reduce the minimum time headway because the following train needs to wait outside the arrival station till the leading train leaves the station and then travel through a safety distance consists of $(\frac{l_1}{2} + \frac{l_2}{2} + d_s)$. To figure out the influencing factors and how they influence the minimum time headway, tests under different safety margins and leading train's dwell time are conducted based on the parameters of case 1. The safety margins and dwell times are set to manually locate the minimum time headway for each case, and the result are shown in Fig. 8. It can be seen that the minimum time headway increases with the increment of safety margin as well as the dwell time. In other words, the time headway can be shortened with the deduction of the safety margin and dwell time. This result inspires the future research on exploring the strategies like multi-platforms or diversion operations which fully utilize the VC operation to help reduce the time headway between each train pairs to enhance the railway capacity.

IV. CONCLUSIONS AND FUTURE WORK

This article establishes a model of VC operation based on the MILP approach and verifies the effect of reducing the train separation on a single rail track with the safety constraints of MBS-RBD. After testing and analyzing the results in three scenarios, it is found that the result of minimizing the train separation leads to a naturally formed VC state after the following train catches up with the preceding one. The process of coupling, coupled, and decoupling under three scenarios with different levels of disturbance verifies the operability of VC.

In the future, the method of model predictive control (MPC) can be introduced to break away from global-scale optimization and increase the operability of the train in actual service. Based on the minimum headway analysis, a multi-platforms station model and rearrangement of timetable can be introduced to further increase the capacity and fully realize the potential of VC operations in future high-speed railway systems.

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